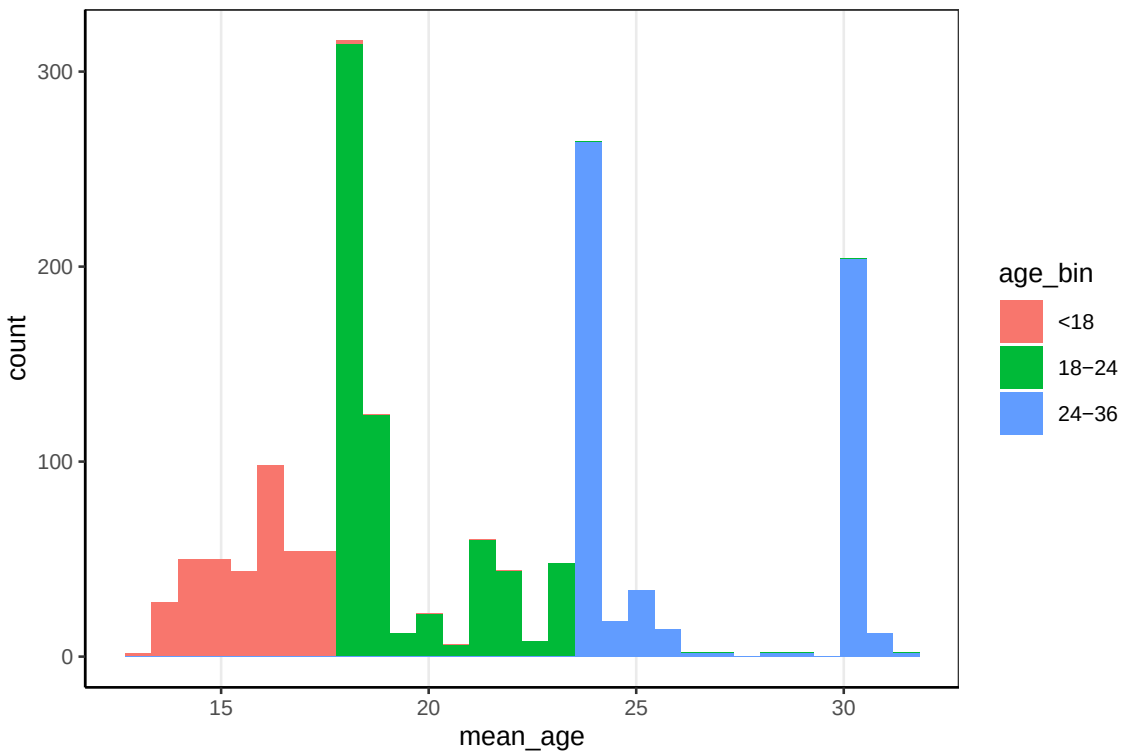
Supplemental Information: TODO TITLE

1 Data description

Figure S1 shows the distribution of ages in the Peekbank data used in this paper.

```
## # A tibble: 1 x 2
##    mean median
##   <dbl>  <dbl>
## 1  22.2     19
```

```
## # A tibble: 1 x 2
##    mean median
##   <dbl>  <dbl>
## 1  25.4   22.6
```



2 Methods details

TODO FIX ## Criteria for inclusion

TODO long version of what makes a trial vanilla (if needed)

2.1 Details about ICC

There are a number of different correlations that all go by the name ICC.

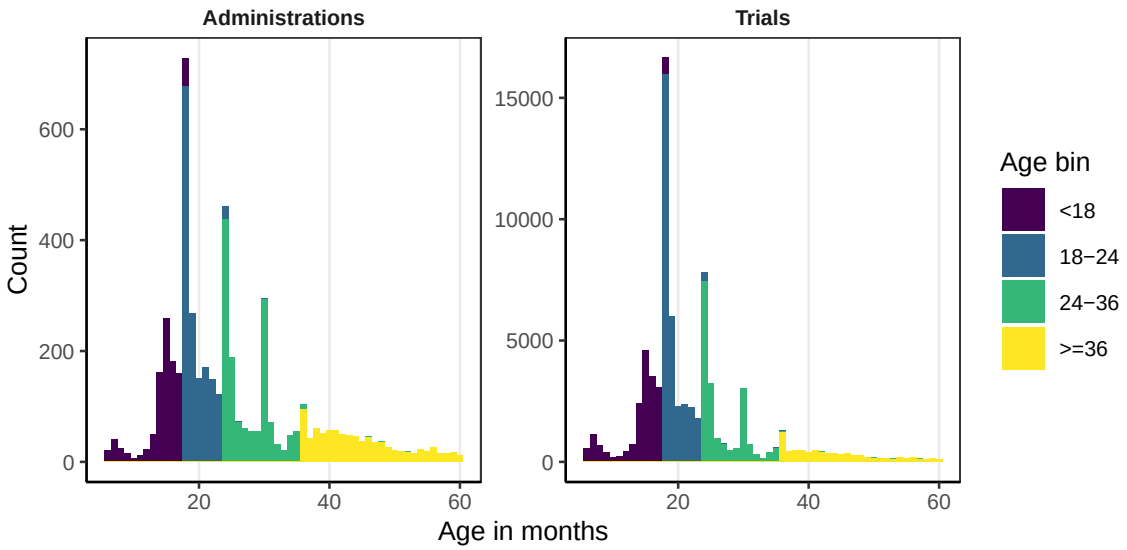


Figure S1: Age distribution for trials and administrations, with the age bins used for analysis.

We are using the ICC for *inter*-rater reliability for a "random factorial model without interaction" measuring consistency and looking for the reliability of an average of raters. This is provided by the "agreement" package we use, but I don't think it is fully documented anywhere.

Basically, this ICC looks at the ratio between by-subject variance (by-subject variance + by-rater variance + error variance)

- inter-rater reliability means we want to know how well raters (i.e.) items correlate with each other (as opposed to with themselves)
- the "random factorial model" (type 2 in standard ICC classification) is for when all raters rate all subjects (not quite true for us), but may have varying numbers of measures on each combination. Both raters and subjects are assumed to be sampled from a larger population of interest.
- the "without interaction" is a variant for when there are (or may frequently be) single ratings per subject and per rater and/or when subject-rater interactions are not of interest (and can be lumped into the overall error). (Using interaction or not can be a bias-variance trade-off; adding the interactions decreases bias but increases variance.) This version is documented in Gwet 2014 (Handbook of Inter-Rater Reliability, page 256) This measures the ratio of by-subject variance / (by-subject variance + by-rater variance + error variance)
- consistency. We aren't interested in exact agreement between raters but rather whether the slopes across raters for different subjects are the same, so it's okay if one rater is consistently lower (i.e. a later-learned word). This omits the by-rater variance from the denominator (I don't know that this is ever explicitly documented for the random factorial model, but it sort of aligns with later discussion of consistency on 272 of Gwet 2014 and is somewhat consistent with https://pmc.ncbi.nlm.nih.gov/articles/PMC4913118/ 's pattern of formulae for different types of ICC)
- average across raters divides the error term by the number of raters (seems vaguely consistent with https://pmc.ncbi.nlm.nih.gov/articles/PMC4913118/)

so, the overall formula for ICC is then by-subject variance / (by-subject variance + error variance / number of raters)

The package "agreement" that we use seems to be one of few that (by using Gwet's formulation) handles cases where there is missing or uneven data (very common in peekbank, especially for post-exclusion or RT data).

The main point is to have an ICC measure can be used for comparison across the same datasets for different methodological questions. We don't try to interpret the ICC against external cutoffs for reliability.

2.2 Handling uncertainty

How do we tell what numerical differences in ICC or in correlations are substantial? The approach we take here is to use bootstrapping which lets us generate per-dataset confidence intervals around the point estimate. We then can use point-estimates and the bootstrapped CIs to calculate a variance-weighted meta-analytic estimate (and CI) to get an overall estimate.

We run separate meta-analytic estimates for each set of parameters, so each parameter's CI includes all the variation from uncertainty over what the "center" of the distribution across datasets is. This is different from the uncertainty we'd get if we stuck the parameters in as fixed effects of the model – there we'd see wide uncertainty about the "intercept" and a narrower estimate for the fixed effect. However, we don't have specific pair-wise comparisons to test, so we stick with running separate models to show, descriptively, the range of meta-analytic uncertainty.

For ICC which is a trial-level measure, we do the bootstrap resampling over trials. For the test-retest and CDI correlations, the correlation used is between the mean value for that administration and another administration/the external CDI.

It turns out that "take the mean across trials then correlate" is a very biased statistic. So, if you do trial-wise bootstrapping on this statistic, the results from the bootstrap samples are not evenly distributed around the sample value, and in fact often are all lower than the sample mean. The easy way to think about this is that sampling trials from the population (bootstrapping) increases the noise on the means-across-trials, and with a noisier measure, the correlation will go down.

Because of this, we do administration-level bootstrapping for the CDI and age correlations and pair-of-admins-level bootstrapping for the test-retest.

3 Accuracy

Fig S2 shows a version of Figure 1 split by age bin categories. Older children have overall higher ICC reliability, but different ages do not show different regions of high reliability.

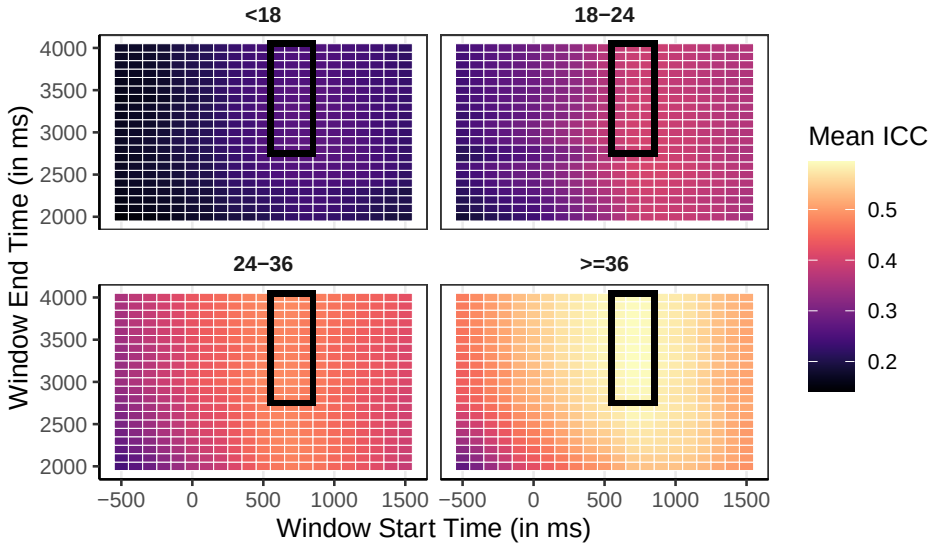


Figure S2: A heat map showing the average ICC across datasets for different start and end points for the accuracy window, split by age bins. A region with the cross-age highest ICC (best reliability) is outlined.

Figure S3 shows the model estimates and 95% CIs for the effect of different windowing options (comparing to 600-4000ms as the index condition). Negative numbers indicate that the value is lower in the given window than for the index condition; higher numbers indicate that the value is higher in the given window than in the index condition.

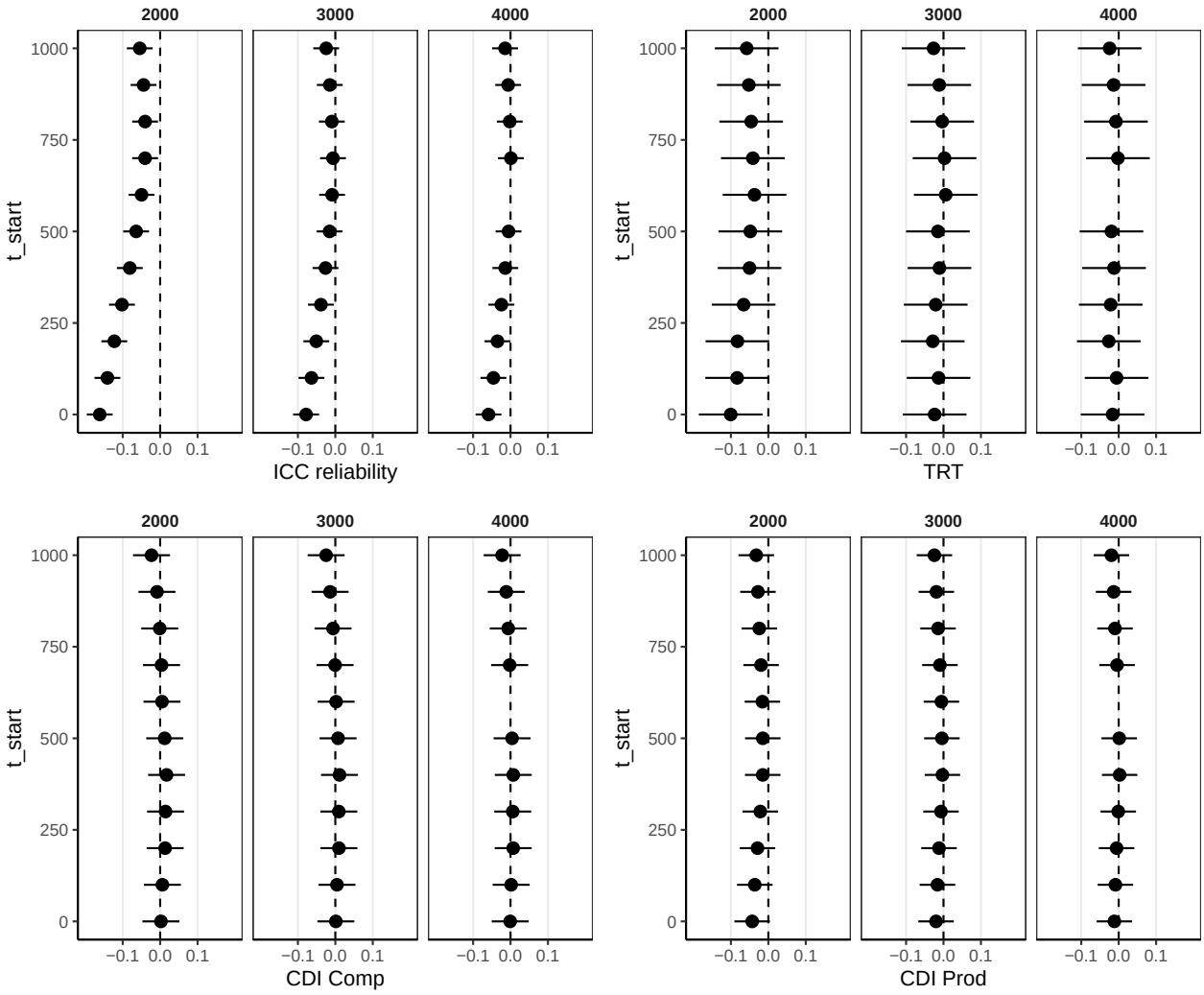


Figure S3: Model results comparing different window lengths to the index length of 600-4000ms.

Figure S4 shows the model coefficients and 95% CIs for linear mixed effect models run separately for each age bin. While the degrees of different vary across ages, the overall patterns are the same.

Figure S5 shows the distribution of AOIs for different datasets and different timepoints. Some datasets have less data at the start or end due to varying length recordings, or being a mix of multiple experiments with different recording windows. Most (but not all) datasets had data before the point of disambiguation and up to 3000ms. Datasets varied dramatically in overall size.

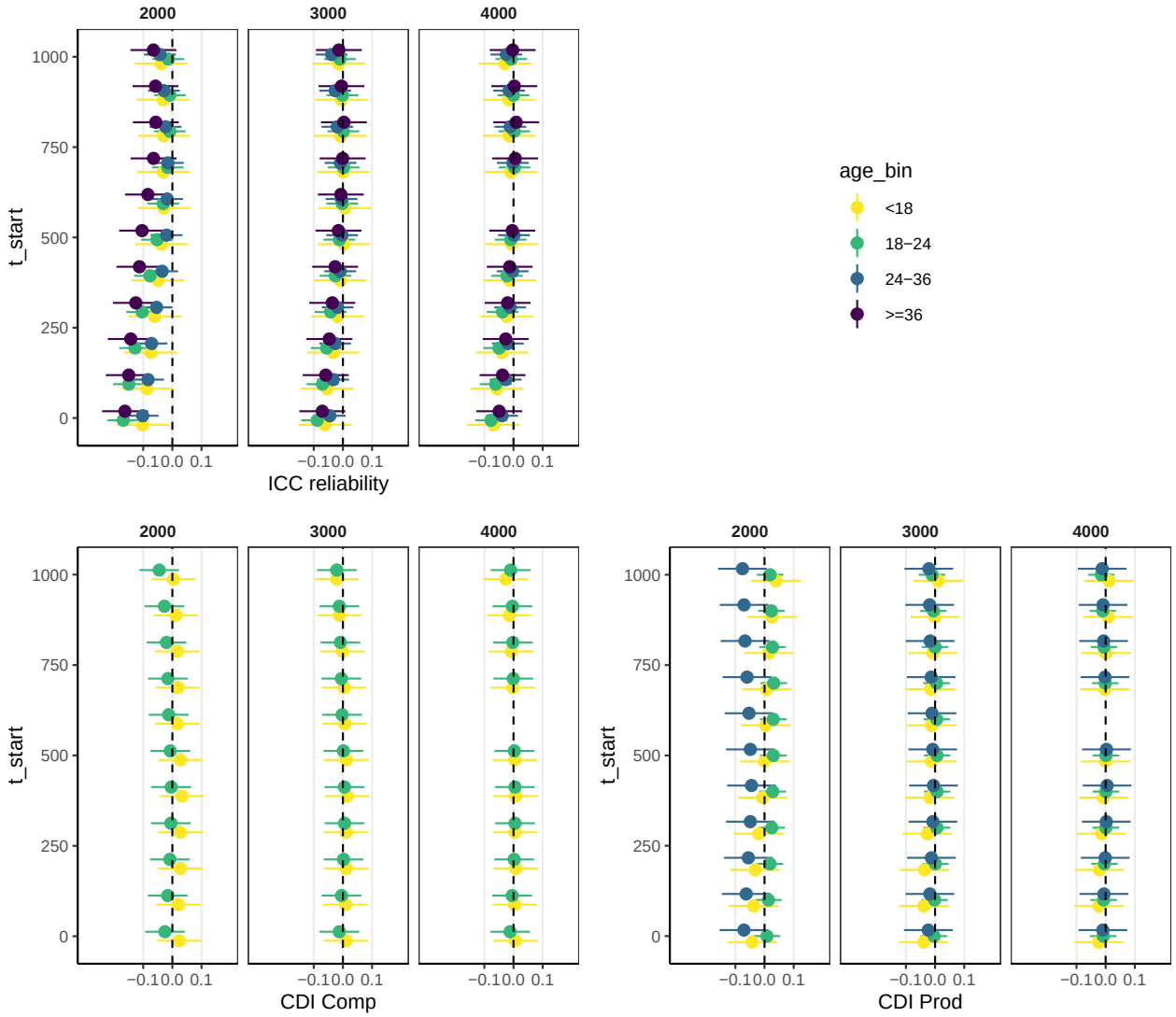


Figure S4: age split models of accuracy, 600-4000 as baseline

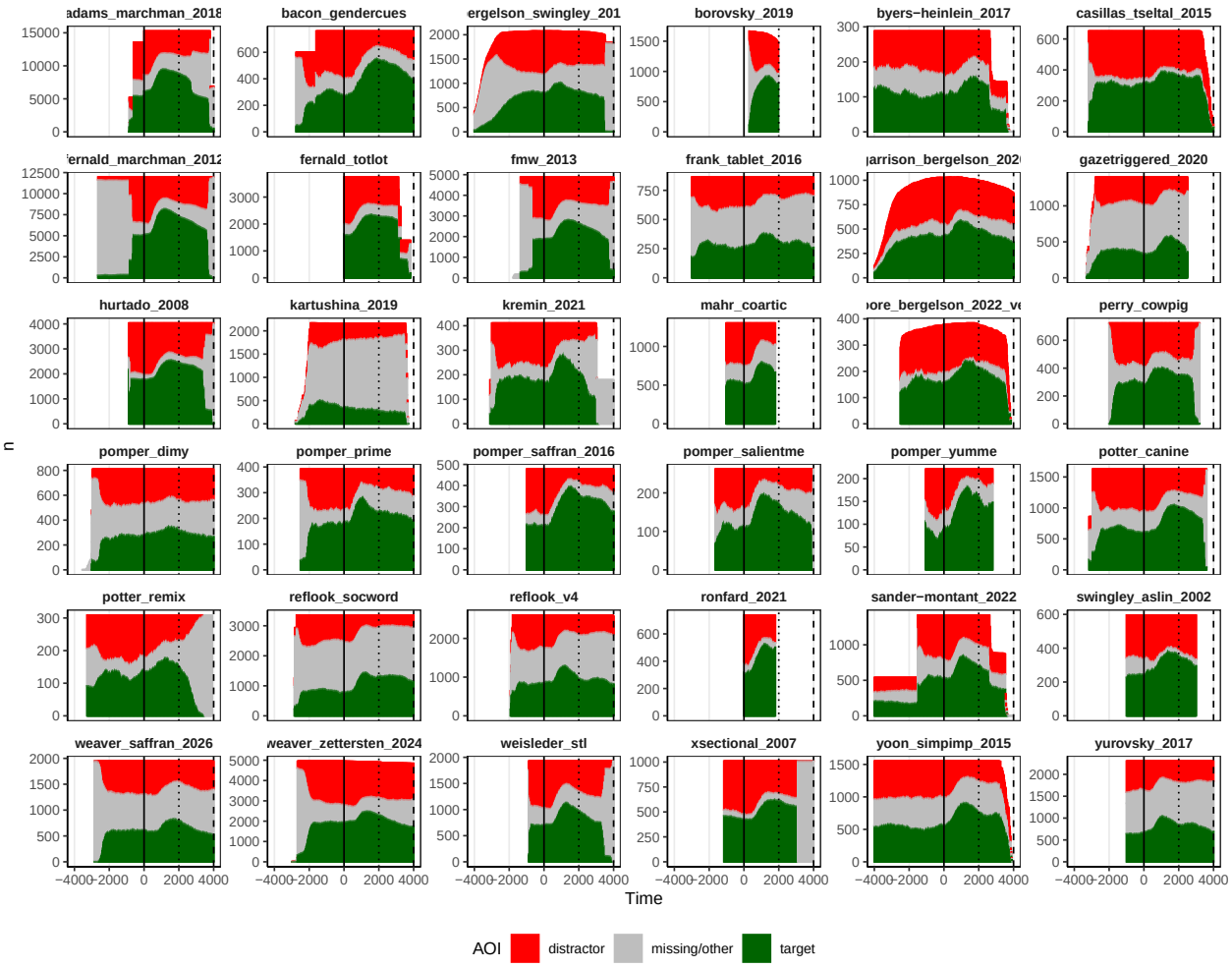


Figure S5: Distribution of AOIs across timepoints and datasets. Solid vertical line at point if disambiguation, Dashed lines at 4000ms, and dotted line at 2000ms.

4 Baseline correction

Figure ?? shows how baseline corrected measures compare to normal accuracy measures for a 200-2000 window, as an example of generalization.

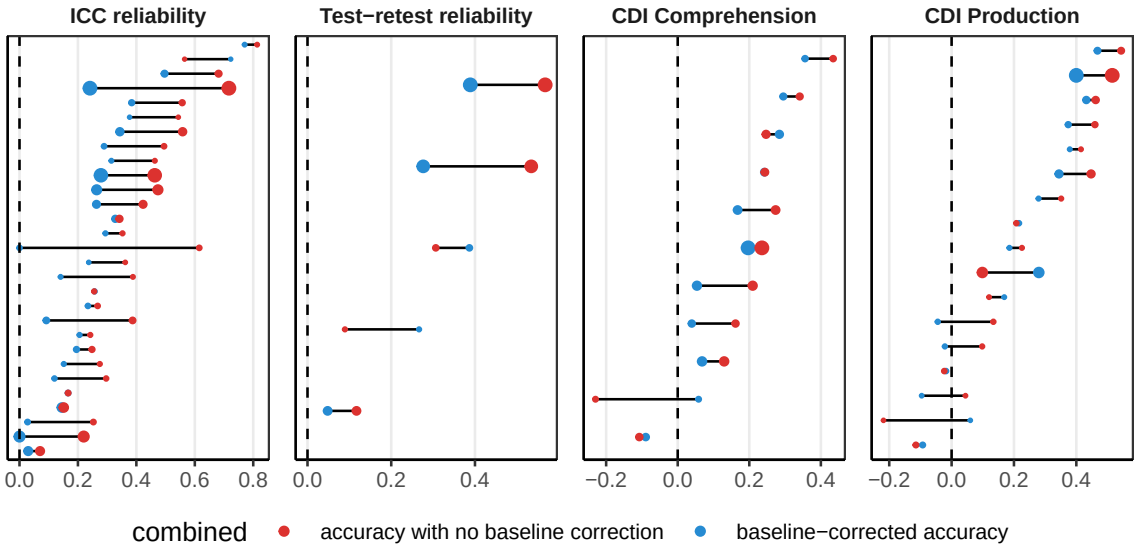


Figure ?? shows model results comparing different baseline correction options to a plain accuracy measure. Baseline correction is substantially worse.

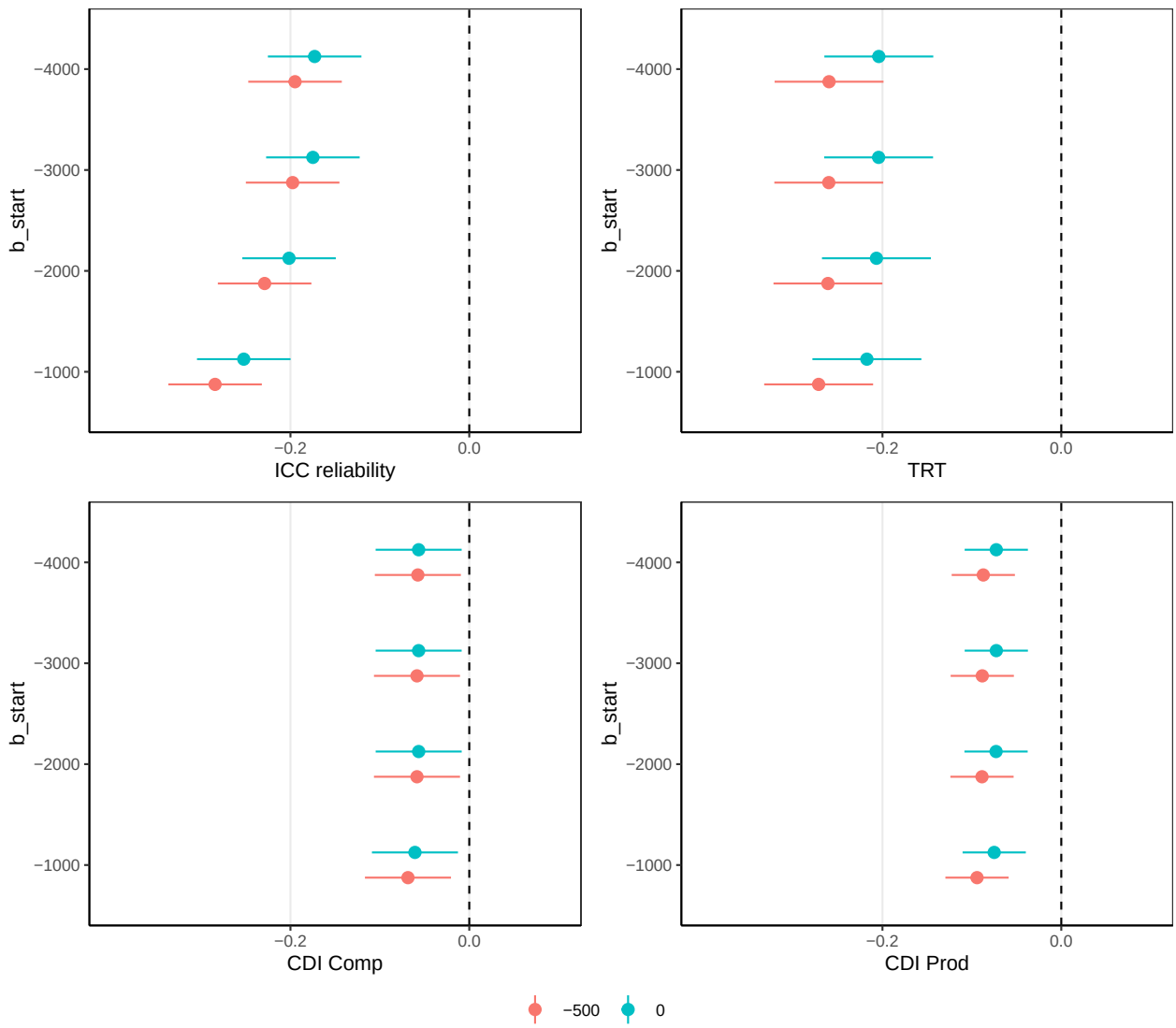


Figure S6 shows model results comparing baseline corrected accuracies to the index condition of normal accuracy for each of the age bins. Across age bins, baseline correction is often worse and in no case better than normal accuracy.

As some datasets have much shorter pre-disambiguation windows than others, we also looked at baseline-corrected accuracy on the subset with data before 3000ms pre-disambiguation (Figure S7). Even on this subset, baseline-corrected accuracies have worse validity and no better reliablity than normal accuracy measures.

5 Reaction Time

Figure S8 shows a heat map of ICC relability across different RT conditions, split by age.

Figure S9 shows models estimates for different model parameters compared to the index condition of untrimmed log land with 400 minimum RT. Because faster RT tends to correlate with higher language skill and thus higher CDI measures, for the bottom two panels, positive numbers indicate worse correlations in the tested condition compared to index.

Figure S10 shows the reliability and validity of different reaction time parameters, split by age bin.

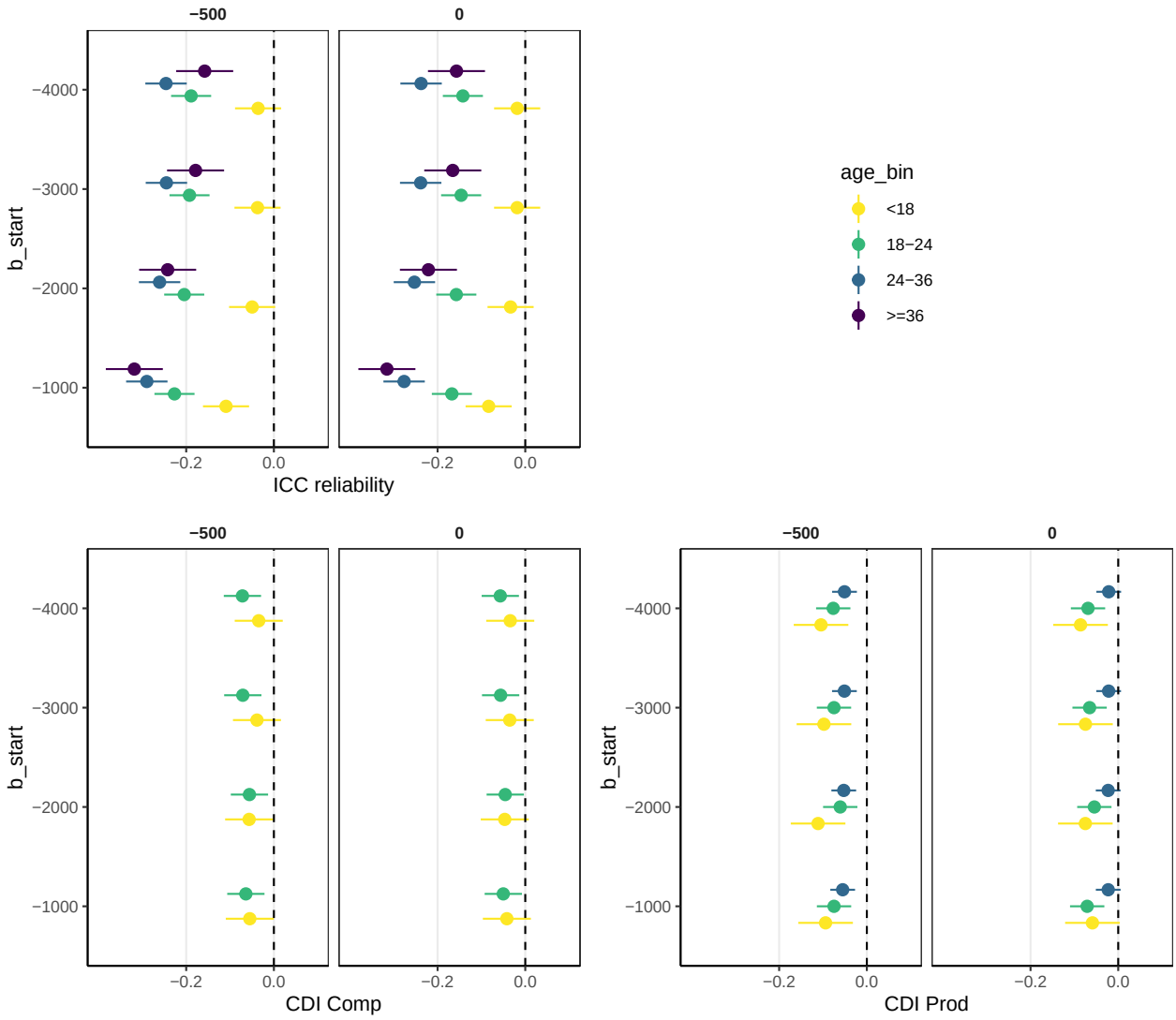


Figure S6: age split baseline correction

6 Exclusions

6.1 trial level accuracy

TODO may need adjustments depending on what realistic range is

kinda weird distribution across different datasets – but might also be a missing data/low data issue?

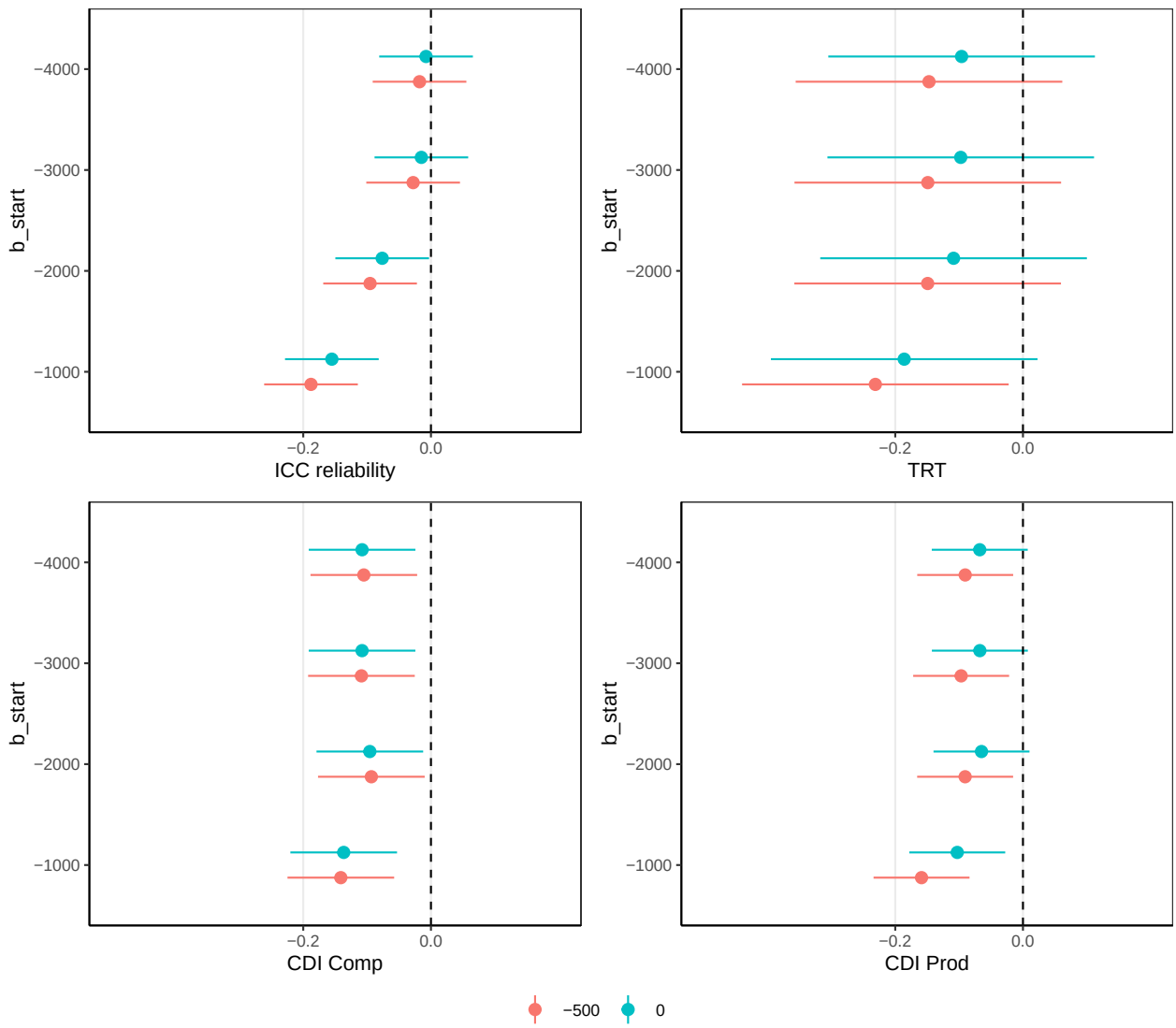
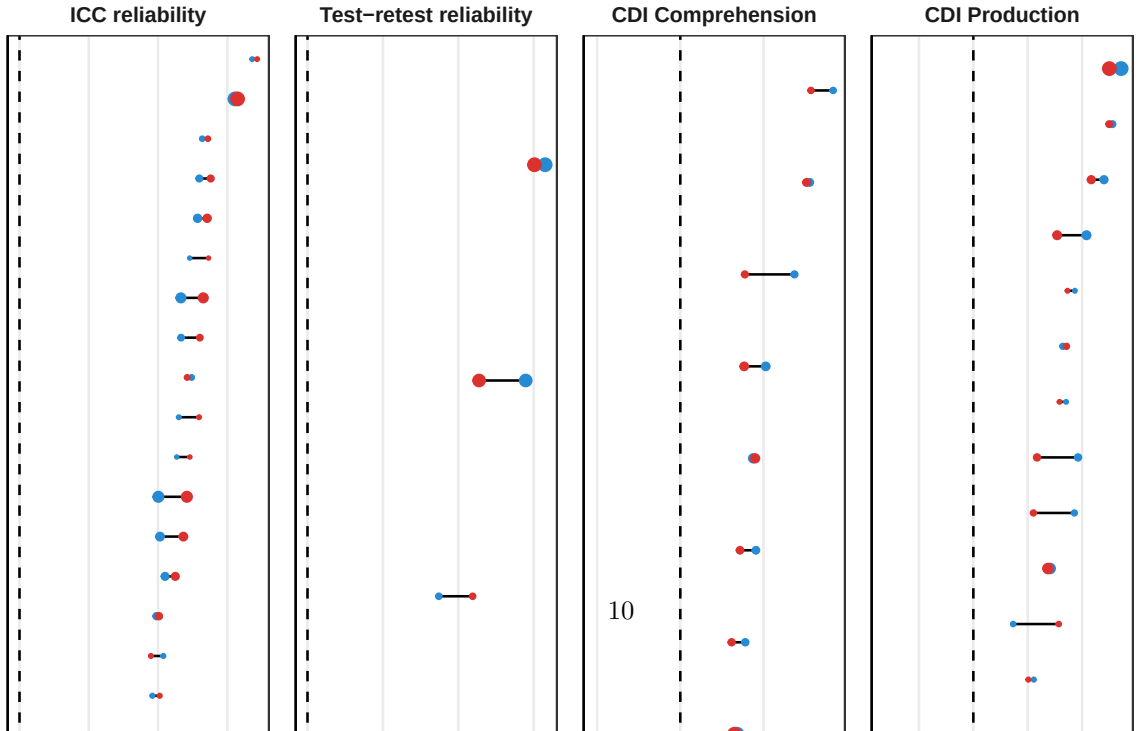


Figure S7: model of baseline correction, compared to no correction



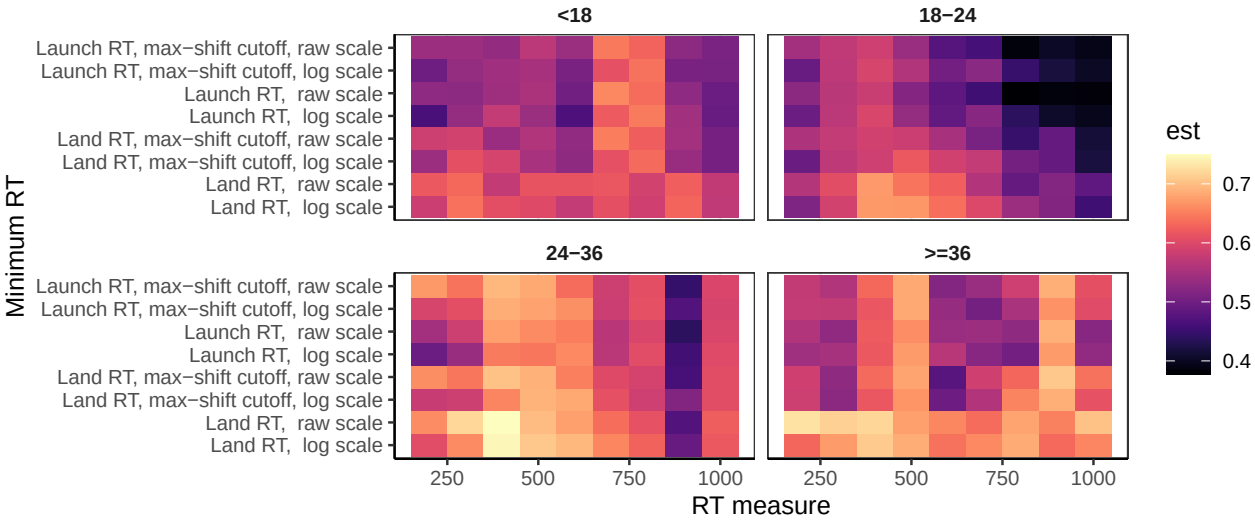


Figure S8: Heat map of different parameter options for measuring RT and how they on average affected ICC.

Figure **??** shows effects of different trial-level exclusions on outcomes. TODO figure out a good way to show the dataloss rates

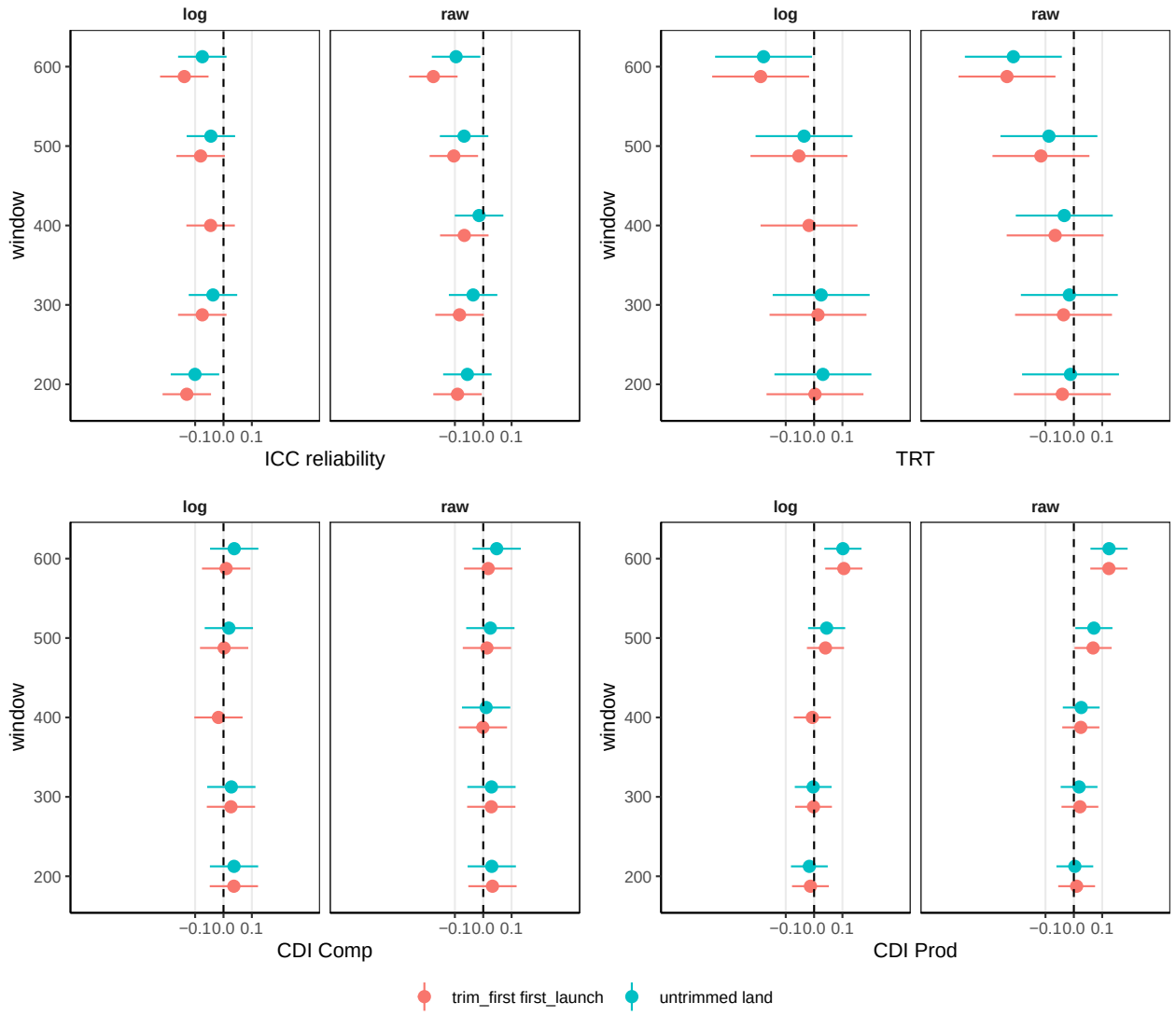
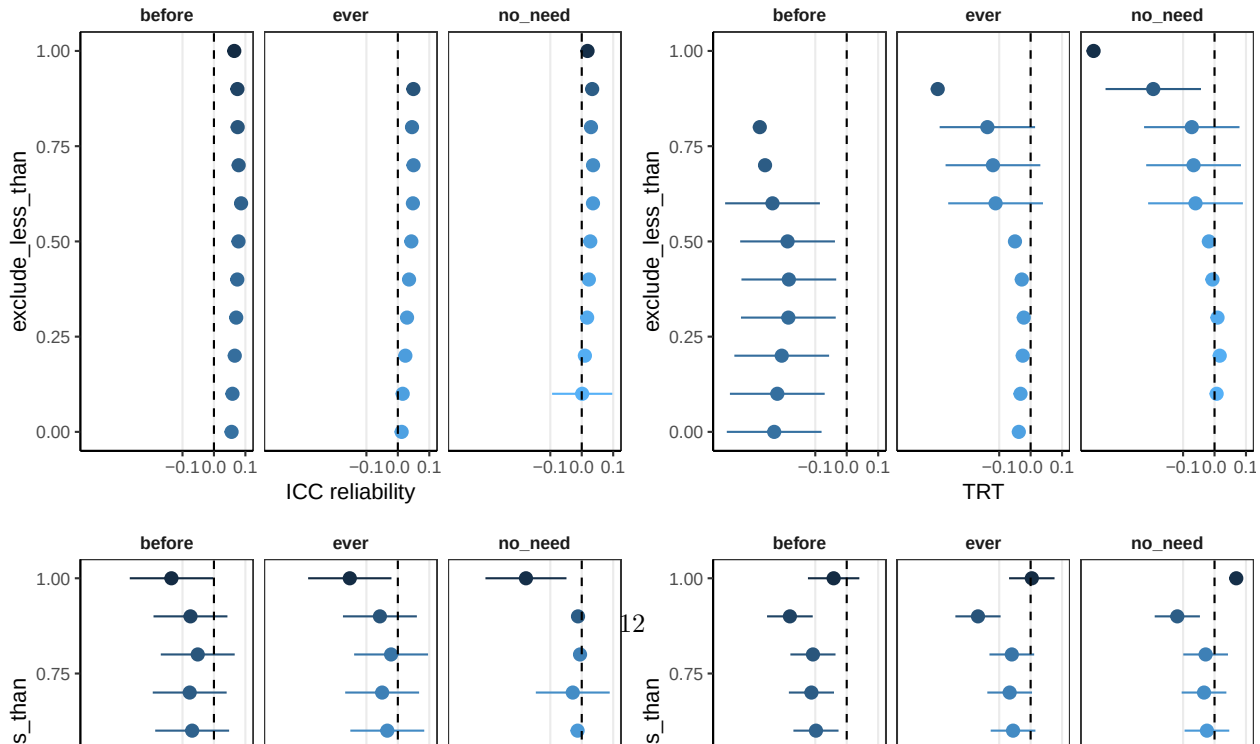


Figure S9: RT, baseline is untrimmed log land with 400 minimum RT



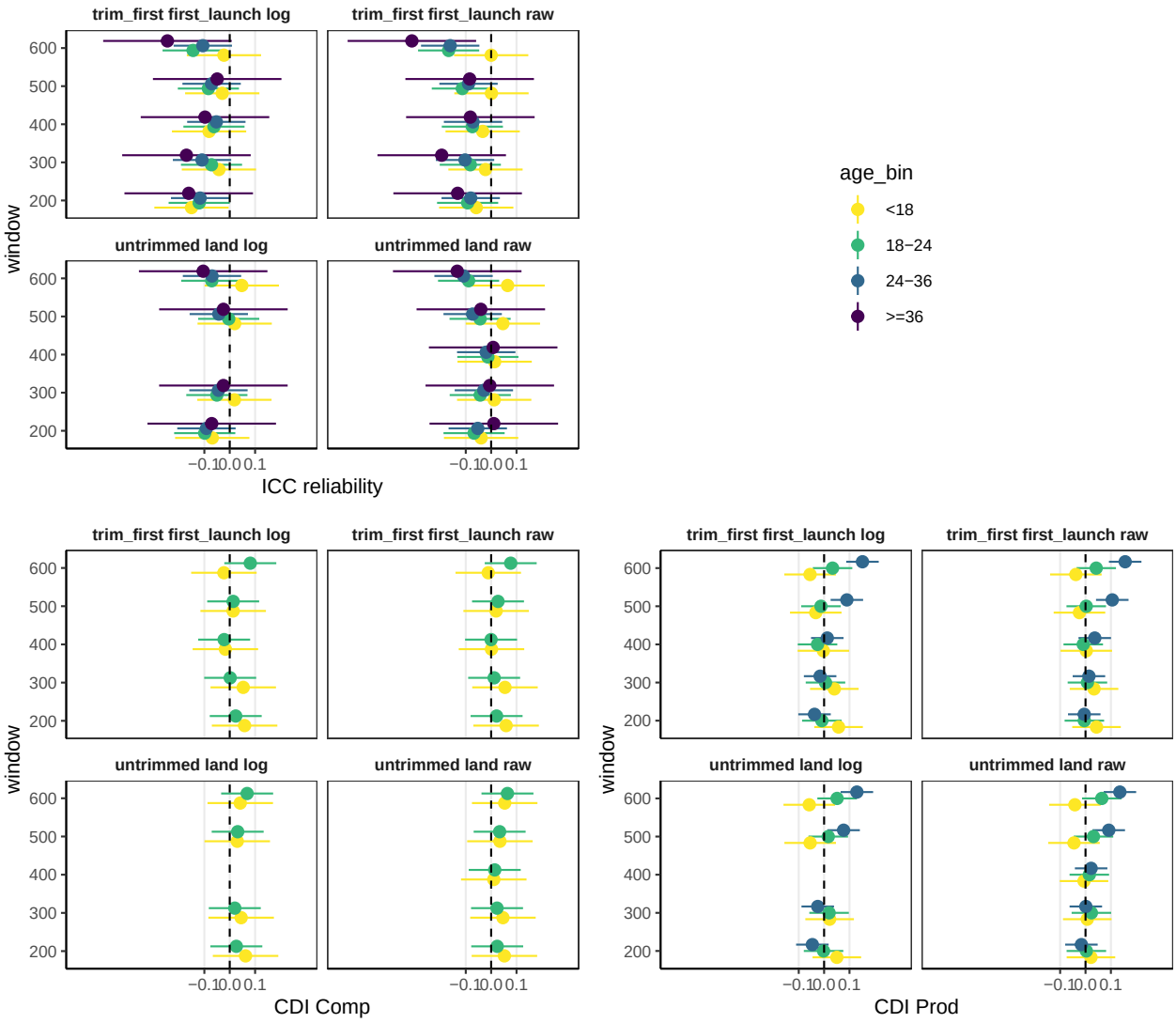


Figure S10: RT age split, baseline is untrimmed log land with 400 minimum RT

Figure S11 shows age splits of the same. TODO HOW TO SHOW DATA LOSS & WHAT ARE REASONABLE BOUNDS

TODO IF NEEDED COULD ADD FOR OTHER ACC WINDOWS

6.2 kid-level accuracy

7 kid-level rt

Figure S12 shows stuff. TODO what sort of min trials do people actually use??

TODO does anyone do kid side bias stuff??

Overall 1.5% of kids (contributing less than 1% of trials) show a bias of 90% or more on more than 75% of trials.

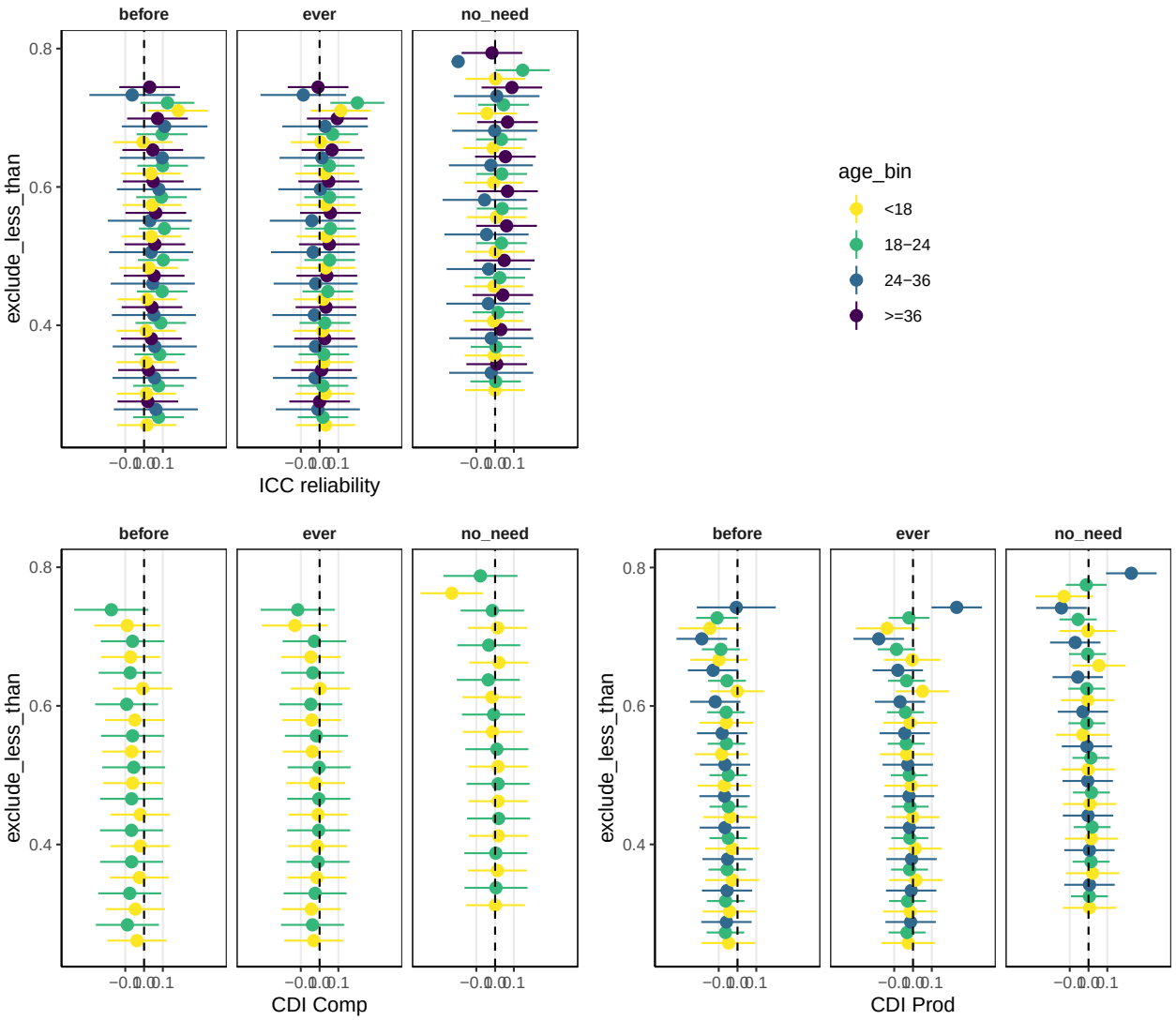


Figure S11: acc trial exclusions

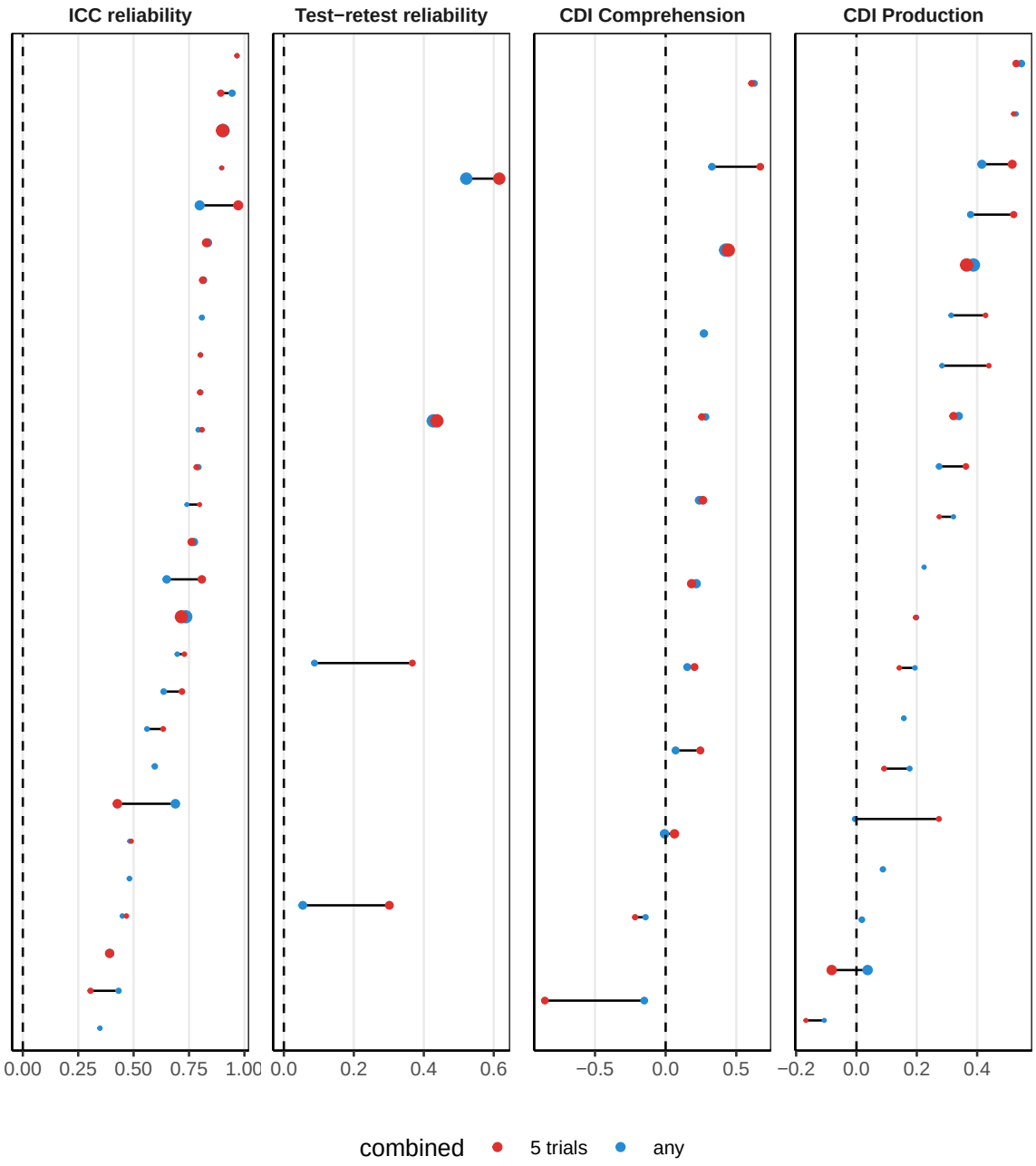


Figure S12: rt, kid level exclusions

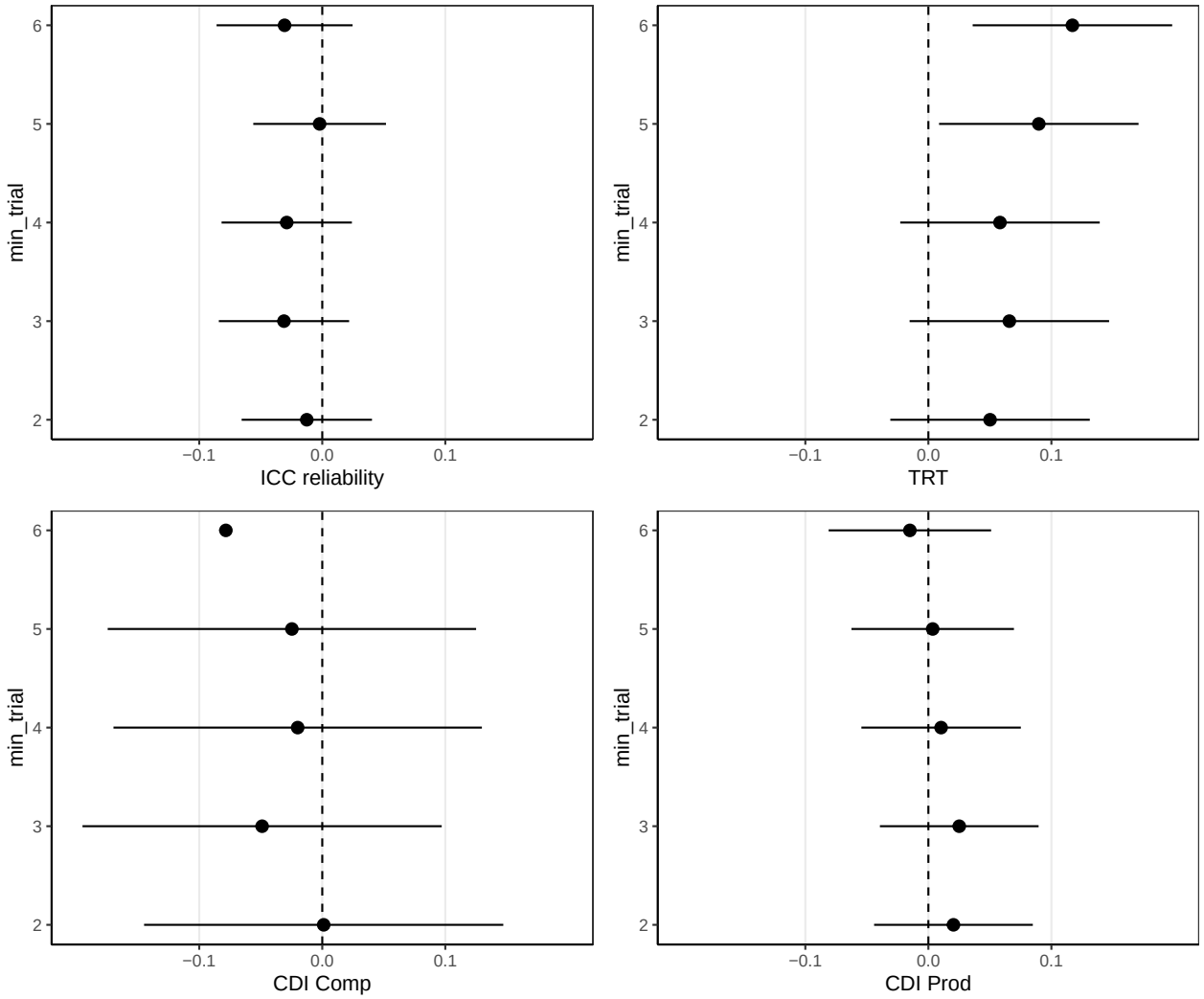


Figure S13: rt kid level exclusions

8 Animacy

Figure S14 shows the time course of animate and inanimate targets and distractors, split by age bin. TODO interpretation

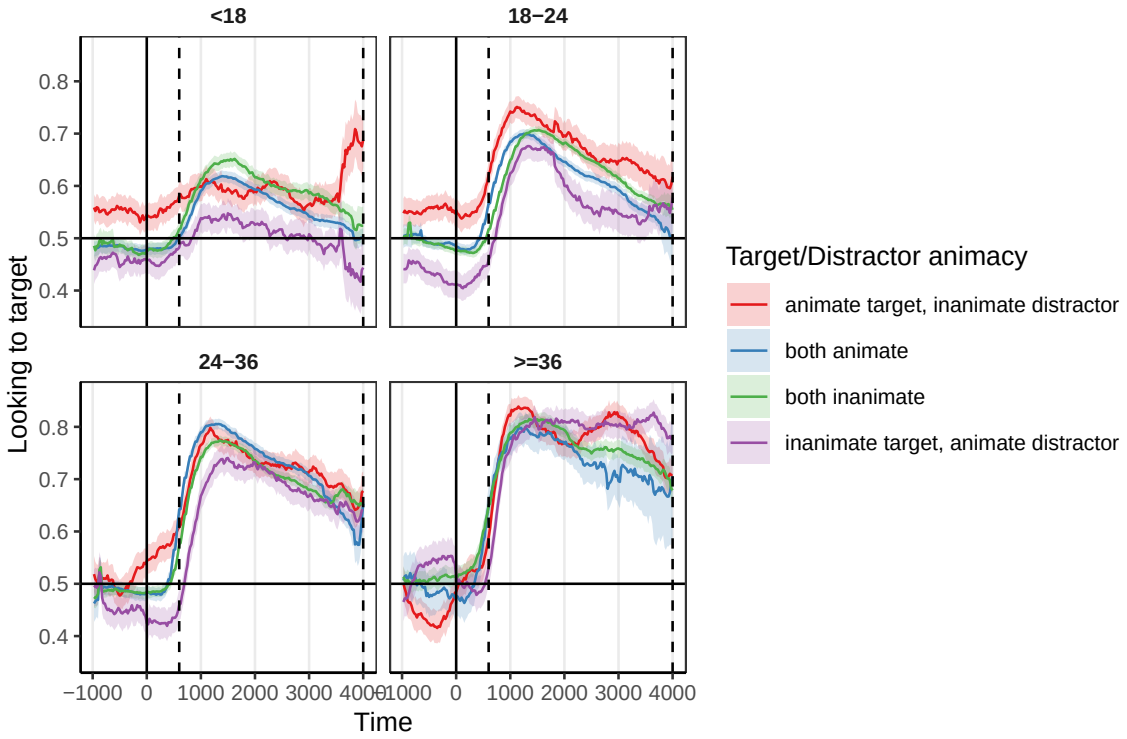
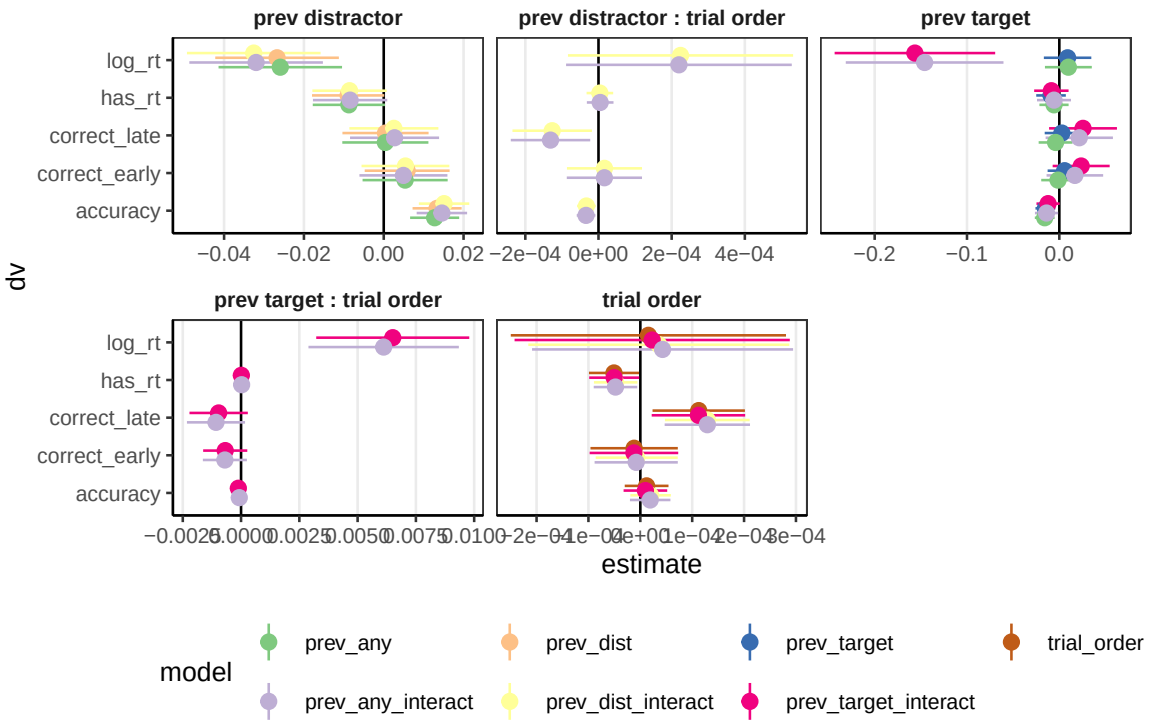


Figure S14: Time course of looking for animate and inanimate targets and distractors, split by age bin.

9 Order Effects

One concern is that TODO

TODO figure out how best to show this!


prev distractor
prev distractor : trial order
prev target
prev target : trial order
trial order
log_rt
has_rt
correct_late
correct_early
accuracy
dv
estimate
model
prev_any
prev_dist
prev_target
trial_order
prev_any_interact
prev_dist_interact
prev_target_interact

10 References